Experiment6

```
!pip install pgmpy
from pgmpy.models import DiscreteBayesianNetwork
from pgmpy.inference import VariableElimination
import pandas as pd
data = pd.DataFrame({
    'Rain': ['No', 'No', 'Yes', 'Yes', 'No', 'Yes', 'Yes', 'No'],
    'TrafficJam': ['Yes', 'Yes', 'Yes', 'No', 'Yes', 'No', 'No', 'No'],
    'ArriveLate': ['Yes', 'Yes', 'Yes', 'No', 'No', 'Yes', 'Yes', 'Yes']
})
model = DiscreteBayesianNetwork([
    ('Rain', 'TrafficJam'),
    ('TrafficJam', 'ArriveLate')
])
model.fit(data)
for cpd in model.get_cpds():
    print(cpd)
inference = VariableElimination(model)
query_result = inference.query(variables=['ArriveLate'],
evidence={'Rain': 'Yes'})
print(query_result)
```

OUTPUT:

```
+-----------+-----+ | Rain(No) | 0.5 | +-----------+-----+ | Rain(Yes)
| 0.5 | +-----------+-----+ +-----------------+----------+-------
----+ | Rain | Rain(No) | Rain(Yes) | +-----------------+----------+--
---------+ | TrafficJam(No) | 0.25 | 0.75 | +-----------------+------
----+-----------+ | TrafficJam(Yes) | 0.75 | 0.25 | +-----------------
+----------+-----------+ +-----------------+----------------+-
----------------+ | TrafficJam | TrafficJam(No) | TrafficJam(Yes) | +----
-------------+----------------+-----------------+ | ArriveLate(No
| 0.25 | 0.25 | +-----------------+----------------+-------------
----+ | ArriveLate(Yes) | 0.75 | 0.75 | +-----------------+-----------
-----+-----------------+ +-----------------+-----------------
-+ | ArriveLate | phi(ArriveLate) |
+=================+===================+ | ArriveLate(No) | 0.2500 |
+-----------------+-------------------+ | ArriveLate(Yes) | 0.7500 |
+-----------------+-------------------+
```